

Aarav Mehta

AI / Machine Learning Developer

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PROFESSIONAL SUMMARY

AI Developer with 3+ years of experience designing, training, and deploying machine learning and deep learning systems, including LLM-powered applications, computer vision pipelines, and production-grade ML APIs. Skilled in Python, PyTorch, and the modern LLM/RAG stack (LangChain, vector databases, prompt engineering). Proven track record of shipping models from research prototype to scalable, monitored production services.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, Bash
ML / DL Frameworks	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Keras
LLM / GenAI	LangChain, LlamaIndex, RAG pipelines, prompt engineering, OpenAI & Anthropic APIs, fine-tuning (LoRA/QLoRA)
MLOps & Infra	Docker, Kubernetes, MLflow, Airflow, AWS SageMaker, GCP Vertex AI, CI/CD
Data & Vector Stores	PostgreSQL, MongoDB, Pinecone, FAISS, Weaviate, Redis
Other	REST/GraphQL APIs, FastAPI, Git, Linux, A/B testing, model evaluation & monitoring

WORK EXPERIENCE

AI Developer — Nimbus Analytics Pvt. Ltd.

Bengaluru, India | June 2023 – Present

- Designed and deployed a Retrieval-Augmented Generation (RAG) chatbot for internal knowledge search, cutting average employee query resolution time by 45%.
- Fine-tuned open-source LLMs (Llama 3, Mistral) using LoRA for domain-specific summarization, improving ROUGE-L scores by 18% over the base model.
- Built and maintained production ML pipelines with Airflow and MLflow, reducing model retraining and deployment time from 3 days to under 4 hours.
- Containerized inference services with Docker/Kubernetes on AWS, serving 500K+ monthly predictions with 99.9% uptime.
- Collaborated with product and data teams to define evaluation metrics and set up automated model-drift monitoring dashboards.

Machine Learning Engineer — Verta Solutions

Pune, India | July 2021 – May 2023

- Developed a computer vision model (YOLOv5 + custom classifier) for automated defect detection on a manufacturing line, reducing manual QA effort by 60%.
- Built end-to-end NLP pipelines for sentiment analysis and support-ticket classification, achieving 92% F1-score in production.
- Optimized model inference latency by 35% through quantization and ONNX conversion for edge deployment.
- Wrote reusable data-preprocessing and feature-engineering libraries adopted across three product teams.

PROJECTS

DocuMind — AI Document Q&A Assistant

- Built a full-stack RAG application using LangChain, FAISS, and FastAPI, enabling natural-language Q&A over uploaded PDFs and Word documents.
- Implemented chunking, embedding, and re-ranking strategies that improved answer relevance by 22% in blind evaluation.
- Tech: Python, LangChain, FAISS, FastAPI, React, OpenAI API

VisionTrack — Real-Time Object Detection Dashboard

- Created a real-time video analytics tool using YOLOv8 to detect and track objects across multiple camera feeds with a live web dashboard.
- Deployed the model as a lightweight microservice using Docker and streamed inference results via WebSockets.
- Tech: Python, PyTorch, YOLOv8, OpenCV, FastAPI, WebSockets

AutoML Benchmarking Toolkit (Open Source)

- Built a CLI tool that benchmarks multiple AutoML frameworks (AutoGluon, H2O, TPOT) on custom tabular datasets and generates comparison reports.
- Published on GitHub with 300+ stars and adopted by several data-science teams for quick model selection.
- Tech: Python, AutoGluon, H2O.ai, TPOT, Pandas, Matplotlib

EDUCATION

B.Tech in Computer Science (Specialization: Artificial Intelligence & Machine Learning)

Indian Institute of Information Technology (IIIT) | 2017 – 2021 | CGPA: 8.7/10

CERTIFICATIONS

- DeepLearning.AI — Deep Learning Specialization
- AWS Certified Machine Learning – Specialty
- Anthropic — Building with Claude / Prompt Engineering for Developers